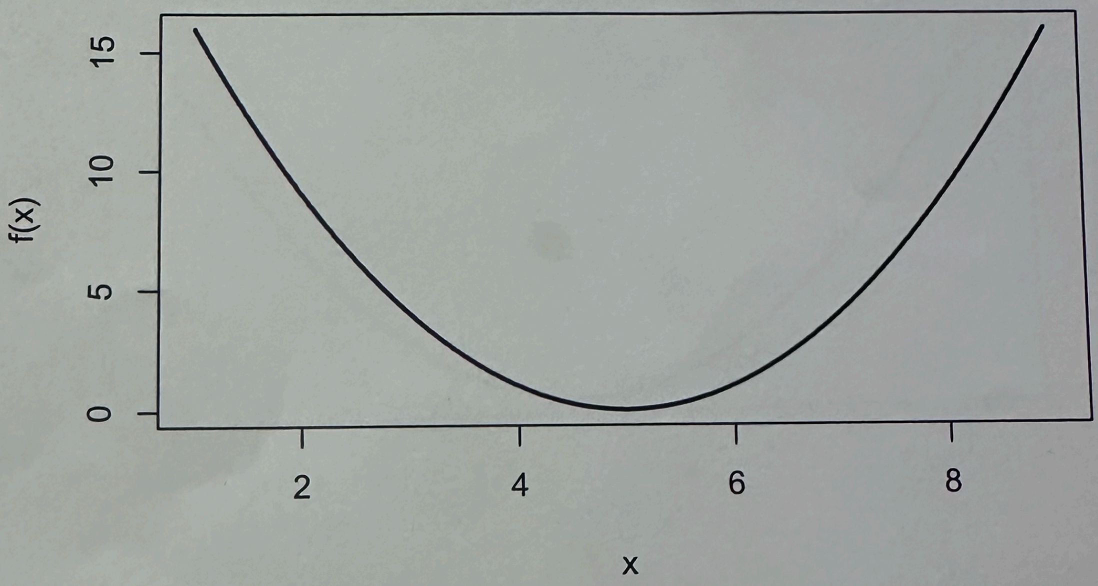
Solution:-

MTH209: Assignment 9

Consider minimizing the function $f(x) = (x-5)^2$.



Starting from 2 and 6 as the two values of the vertices, write down three steps of the Nelder-Mead algorithm:

1. Step 1: $\theta_1 =$ 6, $\theta_2 =$ 2
2. Step 2: $\theta_1 =$ 6, $\theta_2 =$ 4 or $\theta_1 = 4$, $\theta_2 = 6$
3. Step 3: $\theta_1 =$ 5, $\theta_2 =$ 6 or $\theta_1 = 5$, $\theta_2 = 4$
4. Step 4. $\theta_1 =$ 5 $\theta_2 =$ 4.5 or $\theta_1 = 5$, $\theta_2 = 5.5$

ROUGH WORK